

1. PROJECT & TEAM INFORMATION

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models
- **Team ID:** SWTID-2026-7486

2. APPLICATION LOGIC LAYER (BACKEND)

- **Core Language:** Python 3.10+
- **Web Framework:** FastAPI - Selected for its modern, fast, and asynchronous request-handling capabilities.
- **ASGI Server:** Uvicorn - Acts as the lightning-fast web server bridge for running the FastAPI application.
- **Security Module:** python-dotenv - Employed to load protected environment variables and API keys dynamically during runtime.

3. PRESENTATION LAYER (FRONTEND)

- **Dynamic Templating:** Jinja2 - Allows the FastAPI backend to inject generated AI data directly into the HTML structure dynamically.
- **Design & Markup:** HTML5 and CSS3 - Used to craft an intuitive, clean, and responsive user interface without the bloat of heavy JavaScript frameworks.

4. GENERATIVE AI SERVICES

- **Large Language Models (Google):**
 - **Gemini 1.5 Flash:** Engineered into the system specifically for its ability to rapidly output strict, formatted JSON structures (the comic outline).
 - **Gemini 1.5 Pro:** Leveraged for its superior creative writing capabilities to generate engaging dialogue.
- **Text-to-Image Model (Hugging Face):**
 - **Stable Diffusion (v1.5):** Interfaced via API to translate the AI-written scene descriptions into vibrant, high-quality visual art.

5. SUPPORTING PYTHON MODULES

- **PDF Compilation:** `FPDF` - A highly efficient library used to programmatically arrange images and text onto consecutive document pages.
- **Image Processing:** `Pillow (PIL)` - Utilized for local image handling, formatting, and generating emergency fallback visual graphics if network APIs fail.
- **Network Requests:** `urllib.request` - Python's built-in module used to execute secure, dependency-free HTTPS calls to the cloud AI platforms.